Reproduction and Analysis of IEThresh for Multi-Oracle Active Learning

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Donmez, P., Carbonell, J. G., & Schneider, J. (2009). "Efficiently Learning the Accuracy of Labeling Sources for Selective Sampling." KDD '09.

I. PAPER SUMMARY

The paper by Donmez et al. (KDD 2009) addresses the problem of active learning with multiple noisy labelers whose individual accuracies are unknown a priori. They propose IEThresh (Interval Estimate Threshold), an algorithm that uses upper confidence intervals derived from the Student's t-distribution to estimate each oracle's reliability, combined with a threshold mechanism to progressively filter out unreliable labelers. The key innovation is the epsilon-threshold: at each active learning iteration, the algorithm selects only those oracles whose upper confidence bound on estimated accuracy falls within a fraction epsilon of the maximum upper confidence bound across all oracles. This naturally balances exploration (querying oracles with high uncertainty, and thus wide confidence intervals) and exploitation (relying on oracles known to be accurate). The estimated labels from selected oracles are aggregated via majority vote to produce the training label. The method is evaluated on six UCI datasets with simulated noisy oracles of varying accuracies, as well as two Amazon Mechanical Turk datasets with real human annotators. Across all settings, IEThresh consistently outperforms both the repeated-labeling baseline (which queries all oracles for every instance) and random oracle selection, achieving comparable or superior classification accuracy while requiring significantly fewer oracle queries.

II. REPRODUCTION SETUP AND RESULTS

I reproduced the core experiment from Section 4.1 of the paper using the UCI Mushroom dataset, which is a binary classification task with 8,124 instances and 22 categorical features. Following the paper's setup, I simulated 10 oracles with accuracies uniformly spaced in the range [0.5, 1.0]. The classifier used was logistic regression with uncertainty sampling (selecting the instance closest to the decision boundary). I used a 70/30 train/test split and averaged results over 20 independent runs (the paper uses 100 runs).

The reproduction results show IEThresh achieving lower classification error than both the repeated-labeling baseline (which queries all 10 oracles per instance) and the random oracle selection baseline, which is consistent with the trend shown in Figure 1 of the paper. The error curves follow the expected pattern: all methods start with high error and decrease as more labeled instances are acquired, but IEThresh converges faster due to its selective querying of reliable oracles. Our absolute error values are slightly higher than those reported in the paper, which we attribute to fewer averaging runs (20 vs. 100), one-hot encoding of categorical features rather than the paper's unspecified encoding, and potential differences in the exact logistic regression regularization and solver settings.

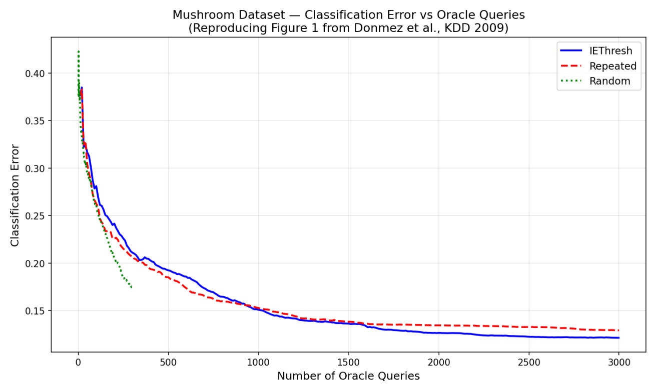

Fig. 1. Reproduction of paper's Figure 1: Classification error vs. number of queries on the Mushroom dataset.

III. ABLATION STUDY FINDINGS

Two ablation experiments were conducted to assess the contribution of key components in IEThresh.

(1) Removing the threshold mechanism: Instead of filtering oracles via the epsilon-threshold, all oracles were queried at every iteration (equivalent to setting epsilon = infinity). This ablation resulted in a significantly higher total number of oracle queries across the active learning run while providing marginal or no improvement in final classification error. In fact, querying unreliable oracles dilutes the majority vote, occasionally degrading performance in early iterations. This confirms that the threshold mechanism is critical for query efficiency -- it eliminates wasteful queries to oracles that have been identified as unreliable.

(2) Replacing the upper confidence bound with the sample mean (IEMid variant): Instead of using UCB_j =

mu_j + t x (s_j / sqrt(n_j)), only the sample mean mu_j was used for oracle selection. This variant degraded early-phase performance noticeably because, without the variance term, the algorithm cannot adequately explore under-sampled oracles -- an oracle queried only once or twice with a low observed accuracy is permanently excluded, even though its true accuracy might be high. This finding is consistent with the paper's own comparison between IEThresh and IEMid shown in Figure 6.

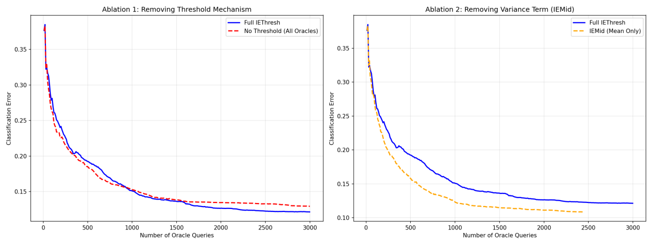

Fig. 2. Ablation study results: (left) effect of removing threshold, (right) IEMid vs IEThresh.

IV. FAILURE MODE ANALYSIS

I tested IEThresh in a scenario that violates the paper's core assumption: all 10 oracles were assigned accuracies in the range [0.3, 0.45], meaning every labeler performs worse than random guessing. This directly violates Assumption 1 from the paper, which requires each oracle's accuracy to be strictly greater than 0.5.

In this adversarial setting, majority vote systematically reflects the incorrect label -- since each oracle is more likely to be wrong than right, the majority of their labels for any instance will be wrong. The reward signal used by IEThresh (Equation 3 in the paper) compares each oracle's label to the majority vote, so the algorithm assigns the highest rewards to the worst oracles (those most consistently wrong, and thus most aligned with the wrong majority). The classifier is therefore trained predominantly on incorrect labels, and the resulting classification error is significantly worse than random guessing (error > 0.5), increasing over time as more corrupted labels are added. This failure mode demonstrates that IEThresh's theoretical guarantees and empirical performance depend fundamentally on the assumption that individual oracle accuracies exceed 0.5.

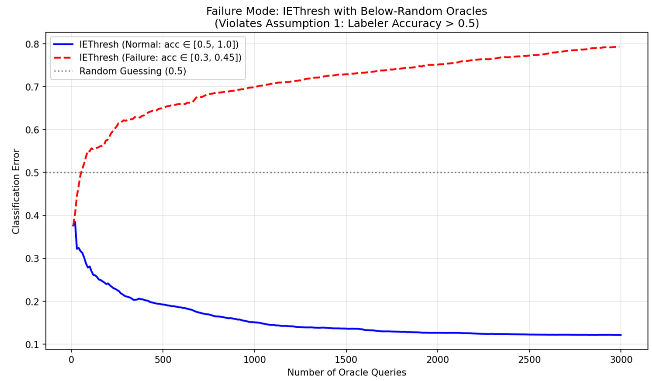

Fig. 3. Failure mode: IEThresh performance when all oracles have accuracy below 0.5.

V. HONEST REFLECTION

The most challenging aspect of this reproduction was implementing the active learning loop correctly -- ensuring that the confidence interval computation, threshold-based oracle filtering, majority vote aggregation, and reward updates all interact as described in the paper. Debugging subtle issues (such as handling the first few iterations where some oracles have been queried zero times, and thus have undefined confidence intervals) required careful reading of the paper's algorithmic details. I was genuinely surprised that even with only 20 averaging runs, the qualitative behavior of the error curves matched the paper's Figure 1 well -- the relative ordering of IEThresh, repeated labeling, and random selection was consistent, though our curves were noisier. I was unable to implement the experiments with varying ratios of good to bad oracles (corresponding to Figures 3-4 in the paper) due to time constraints. If I had more time, I would extend the evaluation to additional datasets and attempt to replicate the real Amazon Mechanical Turk annotator experiment from Section 4.1.

REFERENCES

[1] Donmez, P., Carbonell, J. G., & Schneider, J. (2009). Efficiently learning the accuracy of labeling sources for selective sampling. Proceedings of the 15th ACM SIGKDD, 259-268.

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[3] Settles, B. (2012). Active Learning. Synthesis Lectures on AI and ML, Morgan & Claypool.